

Drying kinetics and energy-exergy performance of a hybrid solar-biomass dryer

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ABSTRACT

Reliable food preservation remains difficult in developing regions with limited electricity and high ambient humidity, where traditional open solar drying often performs poorly. Hybrid solar-biomass dryers offer a promising alternative, yet their energy and exergy behavior for high-moisture vegetables under tropical conditions has not been documented. This study evaluates the drying performance, energy efficiency, and exergy efficiency of a hybrid solar-biomass dryer for cabbage and compares it with standalone sunlight and biomass heating. Experiments were conducted with 4 kg, 8 kg, and 12 kg cabbage loads under three heating modes. Temperatures, solar radiation, relative humidity, and moisture content were measured periodically, and thermodynamic performance was assessed using a validated energy-exergy model implemented in Engineering Equation Solver. Drying durations were 13 h, 8 h, and 7 h for the 4 kg sunlight, biomass, and hybrid modes; 19 h, 13 h, and 12 h for the 8 kg batch; and 20 h, 17 h, and 12 h for the 12 kg batch. The hybrid 12 kg test achieved the highest moisture evaporation rate (766.15 g/h). For the 8 kg trials, energy efficiencies were 38%, 33%, and 50%, and exergy efficiencies were 37%, 33%, and 45% for sunlight, biomass, and hybrid modes, with corresponding exergy destruction of 33.91, 167.54, and 107.42 MJ. By combining stable heat supply with reduced humidity, the hybrid system markedly improved drying rate and thermodynamic performance, demonstrating strong potential for scalable, low-cost food drying in off-grid tropical environments.

Introduction

Food security is a significant concern in developing countries, which sometimes occurs due to a lack of proper preservation measures rather than a shortage of crops. This usually results in either an excess or a shortage. Even though the open sun drying process has always played a crucial role in crop preservation from a very old age, it has some major drawbacks [1,2]. For the safe storage of food or crops, many sophisticated drying methods have been developed over time. These drying processes remove the moisture content from the sample by flowing hot, dry air over it [3–5]. The sample's equilibrium water content determines the process's duration, and it lasts until food saturation pressure equals ambient pressure. Low equilibrium moisture content is preferred in high relative humidity climate conditions because it prevents food contamination from the excessive moisture [6,7]. During the process, the product's outermost layers evaporate moisture as the surface temperature increases. The water content slowly rises to the outer layer due to

diffusion. Water evaporates when the surface moisture vapor pressure exceeds the ambient air evaporation pressure. The moisture removal rate sets the length of the drying process, which can be influenced by different factors, including the types of products and their initial moisture level. The product's heat conductivity can also affect the process sometimes when the drying membrane is thick. The drying time is determined by the rate of moisture diffusion within the food layers and the water evaporation rate from the outer layer for slow and rapid moisture removal rates, respectively. This is crucial for understanding when the drying process effectively commences.

Solar drying technology is a very old and traditional drying method. It's an excellent and prospective solar energy utilization. It's widely used in the agriculture sector since solar radiation aids in direct drying. As the outer membrane absorbs radiation directly, it improves heat transfer by raising the product's inner and outside temperatures. Although the product's radiation absorption capability influences surface heating, most agricultural crops exhibit relatively low solar absorptivity [6]. Moreover, directly drying under the sun is often not that effective, as it

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Nomenclatures			
C_p	Specific Heat	a	Air
e	Specific Exergy	av	Average
$\dot{E}$	Exergy Flow Rate	d	Destruction
h	Specific Enthalpy	ev	Evaporation
s	Specific Entropy	f	Saturated Liquid State
$\dot{m}$	Mass Flow Rate	g	Saturated Vapor
P	Pressure	l	Loss
$\dot{Q}$	Total Heat Transfer Rate	p	Product
R	Gas Constant	v	Vapor
T	Temperature	w	water
v	Specific Volume	q	Specific Heat
P_v	Vapor Pressure	1	Dryer Inlet
P_g	Water Saturation Pressure	2	Dryer Mid-Section
x	Mole Fraction	3	Dryer Outlet
RH	Relative Humidity		
Subscripts		Symbols	
0	Ambient Condition	η	Efficiency
		ω	Humidity Ratio

takes a very long time to achieve the desired moisture content on the food, and sometimes reduces the quality. So, solar dryers are built with several components to obtain the desired efficiency and quality. The main function of the solar dryers is to generate a temperature higher than the ambient environment. As the food gets heated above ambient temperature, the moisture's vapor pressure rises, and it starts to evaporate from the sample. Air absorbs this moisture. This air flow is generated by either natural convection or forced convection using a motor fan. Higher temperature lowers the relative humidity and increases the heated air's ability to absorb water from the crop sample [3–6]. Depending on the component design, different operating parameters, air flow modes, heat generation and transfer modes, solar dryers can be classified into various types [8]. Based on the process of heat generation using solar irradiance, solar dryers can be categorized into direct, indirect, mixed, and hybrid types [9,10].

In direct solar dryers, the sample is exposed to the sunlight inside the drying chamber, which is wrapped in a transparent material to create shade [11,12]. In favorable weather conditions, the temperature inside the chamber can reach high enough to dry various foods, including fruits, vegetables, meats, and spices. But it's very challenging to maintain the desired food quality as there is no temperature control system. This is a major drawback of this system and makes it inappropriate for large-scale applications [13,14]. Despite these drawbacks, the dried sample can maintain good quality for several months when packaging is done properly [15,16]. Arrangement is slightly different for indirect solar dryers [17]. Solar collector utilizes the solar irradiance to warm the air circulating above it, which is directed to the drying chamber to absorb moisture through convective heat transfer. It's been observed that the dryer performs more effectively and efficiently when forced convection is used [18–20]. But appropriate air flow is critical as the air's moisture pickup efficiency reduces with increasing mass flow rate [21,22]. When equipped with PV panel-powered inlet fans, the performance and efficiency have been found to surpass traditional ones [23,24]. However, chimney-type dryers showed better performances [25]. Integration of Phase Change Materials (PCMs), heat storage systems, mesh absorbers, and thermal energy storage units substantially reduces drying time and improves efficiency and food quality [26–32]. Inserting fins in thermal storage units further improves the drying performance [33].

Mixed-type solar dryers combine the principles and benefits of both direct and indirect solar dryers. This system enables a clear drying compartment with air preheating capability [34]. Studies have shown

that this arrangement ensures better thermal utilization, hence increasing the overall drying performance and energy-exergy efficiency compared to direct, indirect, and open sun drying [35,36]. Direct sun radiation improves the initial heat input and drying process. Translucent or transparent materials like glass or plastic coverings trap solar heat, causing a greenhouse effect which raises the air temperature and facilitates convective heat transfer. Indirect solar energy is utilized through a heat exchanger system. A heat transfer fluid absorbs solar irradiance and gets to a higher temperature inside tubes or channels, which is directed to the chamber to transfer heat. The dryer can be customized for specific drying needs by adjusting the ratio of indirect and direct solar radiation, controlling the temperature and airflow, thereby offering more versatility when drying various product types with differing moisture contents and heat sensitivities [35–37]. Integrating thermal energy storage further makes the drying process uniform and ensures product quality [28–38].

Hybrid dryers use other sources of energy, such as biomass combustion or grid electricity, to provide continuous drying assistance when solar power is not enough or not available at all [39–41]. It can be easily integrated with indirect and mixed-type dryers, making it viable for small-scale and larger commercial applications [42]. It been observed to maintain better food quality and hygiene compared to open sun drying [43–45]. The ability to incorporate another auxiliary source of energy, including renewable options, makes it more feasible for rural areas to improve both energy utilization and food preservation [46–48]. Advanced features such as automatic heating, forced convection, and smart control of temperature and humidity enhance the performance to ensure consistent drying and high product quality [49,50]. Use of photovoltaic thermal air collectors and greenhouse structures reduces the biomass consumption and improves thermal efficiency [51–53]. Even though forced convection generally improves the drying rate, it has been found that natural convection is more energy efficient than forced convection in certain setups [54]. Furthermore, integration of thermal storage [55], Phase Change Materials (PCMs) [56], especially nano-enhanced types improve thermal storage and results in faster drying cycle [57–59]. This can be increased by combining PCM chambers with infrared heaters [60]. PCM improves collector performance in flat-plate systems as well and can strengthen thermo-economic, exergy outputs of the system when integrated with parabolic trough collectors [61–62]. Such parabolic dish dryers boost drying efficiency significantly [63]. Similar benefits are observed in cabinet dryers, where optimized PCM placement reduces drying time and improves thermal performance [64],

while air-recycling mechanisms combined with PCM further increase drying and exergy efficiencies in parabolic trough-assisted dryers [65].

There are some major challenges despite all these advantages. Their performance depends on the seamless integration of multiple heat sources and a reliable control system for managing heat input and airflow. This complicates the construction and leads to higher manufacturing and operating costs. Uniform drying is another concern that directly affects the product quality and sustainability [66,67]. Although continuous research has improved the drying technology significantly, there is still a clear need for effective, efficient, and low-cost hybrid drying systems suitable for high-humidity tropical regions.

The present study investigates the energy and exergy performance of a hybrid solar-biomass dryer for cabbage drying under hot-humid tropical operation. Although hybrid solar-biomass drying has been widely reported, a clear gap remains in the availability of design and deployment-ready thermodynamic evidence that is rigorously supported by instrument calibration, uncertainty analysis, and reproducibility assessment, particularly for high-humidity climates. This study bridges that gap by presenting a systematic thermodynamic and sustainability assessment of a heat-exchanger-based, triple-mode solar-biomass tunnel dryer operated in solar-only, biomass-only, and hybrid configurations across multiple drying loads under controlled loading density, rather than reporting a single-condition performance demonstration. The

comparative mode-load framework enables meaningful load-configuration performance mapping for operational planning and implementation. Beyond first-law indicators, the study adds novelty by incorporating a second-law (exergy) diagnosis that identifies dominant irreversibilities across operating modes and translates them into actionable design and operating implications. In addition, the work connects thermodynamic behavior with practical adoption metrics by integrating economic evaluation and environmental impact (carbon-emissions) comparison tailored to Bangladesh-like tropical contexts, which translates performance differences into cost and sustainability trade-offs relevant to rural and semi-industrial use. Moreover, the study provides a transferable kinetic description by evaluating thin-layer drying models and selecting a robust predictive form for cabbage drying suitable for engineering design and simulation.

The primary objective of this study is to evaluate the energy and exergy performance of a hybrid solar dryer for cabbage drying in hot-humid tropical regions. This study provides a very detailed performance evaluation of hybrid solar drying of cabbage under such conditions, offering new experimental data and insights essential for scaling such technologies for rural and semi-industrial applications. The results reveal important trends in moisture diffusion, heat utilization, and exergy destruction, identifying potential areas for improving hybrid dryer design and operational strategies. These findings contribute to the

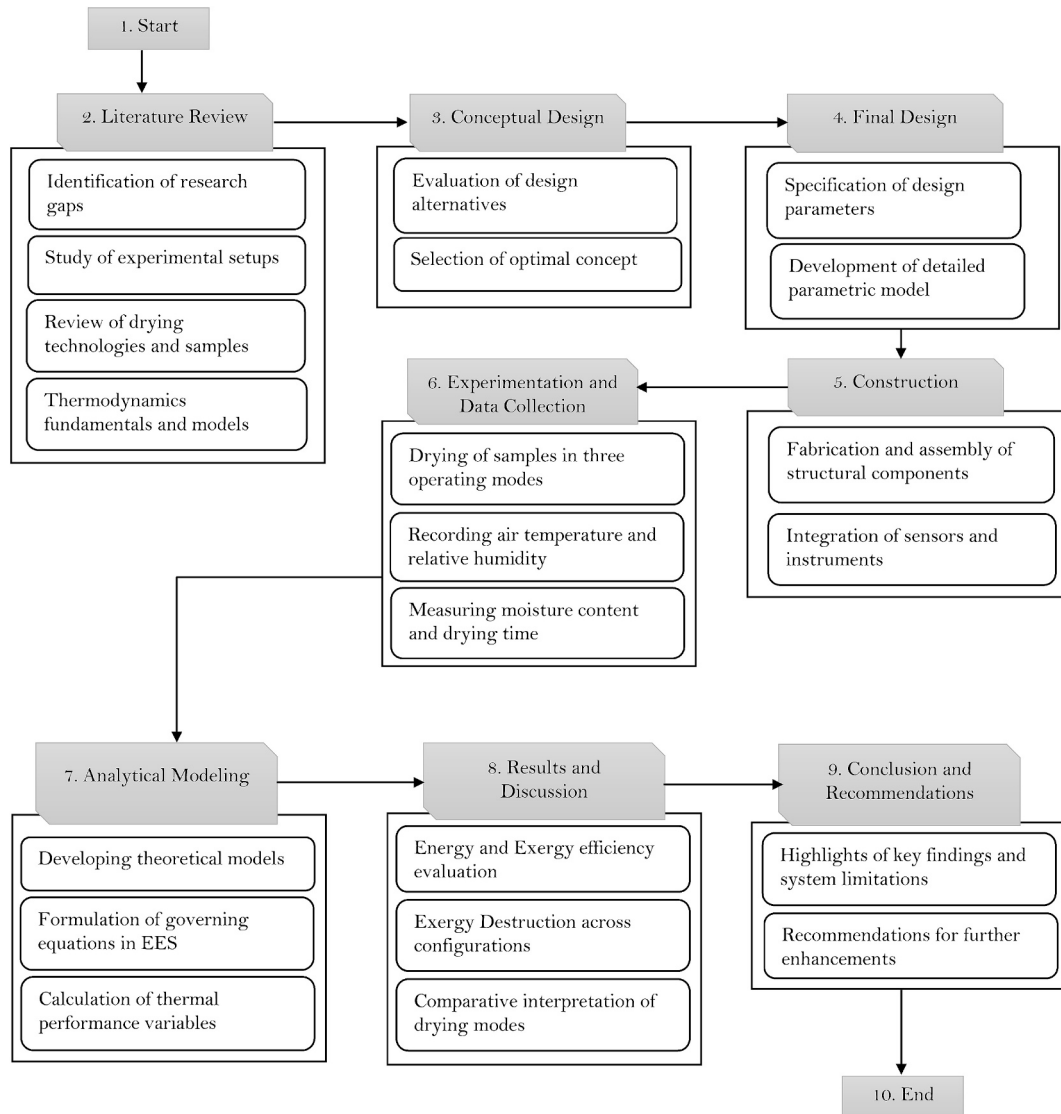


Fig. 1. Workflow Diagram.

development of cost-effective, locally adaptable drying technologies that can support food preservation, reduce post-harvest losses, and guide future advancement toward sustainable, IR 4.0-compatible solar drying solutions.

Methodology

The research followed the steps shown in Fig. 1, similar to processes followed by other reference experimental studies [55–68]. The study began with understanding the problem statement regarding food processing and limitations of conventional drying technologies after a rigorous literature review. The hybrid solar dryer was constructed according to the requirements and the test was conducted using cabbage sample as depicted in Fig. 2. Various parameters related to the process such as ambient temperature and pressure, RH, temperature of inlet, drying compartment and outlet, initial moisture content and final moisture content of the product, air velocity, solar intensity and total drying time was recorded during the experiment. Mathematical models were developed for different drying configurations that were investigated in this study based on thermodynamic principles and by studying various journals and books related to particular analyses. Various calculations were performed based on the derived governing equations in order to evaluate the drying performance and efficiencies of different drying modes using Engineering Equation Solver (EES). The obtained results have been presented to understand the current states and room for improvement. A comprehensive discussion on the drying process has been provided with a comparative analysis of the performance. The study concluded with future recommendations for improved and efficient technology after discussing its possibility for adaptation to the fourth industrial revolution.

Study site

Experiments are conducted at Gazipur Agricultural University (GAU), Bangladesh (24.0362° North, 90.3959° East), a region characterized by high relative humidity typical of tropical climates. Ambient dry-bulb temperature, relative humidity, and solar irradiance are recorded throughout each run. The reference environment for exergy

analysis is taken as the measured ambient state at each time step.

Experimental setup

The overall system design was guided by insights from previous literature on solar and hybrid drying technologies. Prior studies highlighted the importance of optimizing solar collector dimensions for adequate solar irradiance capture [69], adopting forced convection to enhance drying rates, maintaining a greenhouse environment within the chamber to accelerate moisture removal, and integrating heat exchanger units to recover biomass flue gas efficiently. Baffle arrangements were also emphasized for achieving uniform heat distribution, and proper exhaust structures were recommended for safe release of combustion gases.

Building on these established principles, the present study incorporated several novel and application-specific design features. An efficient shell-and-tube heat exchanger was developed to maximize heat transfer from the biomass source while ensuring stable thermal delivery to the drying chamber. A dedicated exhaust system was engineered to improve environmental safety and reduce operational emissions, supporting sustainable operation. Furthermore, the drying chamber geometry and airflow pathways were customized specifically for cabbage, a high-moisture leafy vegetable with unique structural and thermal characteristics that differ from commonly studied grains or spices. These design refinements were targeted to enable more uniform heating, improved moisture diffusion, and enhanced system performance under targeted climate conditions. The designed model and fabricated system have been presented in Fig. 3.

The drying chamber was a semi-circular tunnel covered with a transparent polythene sheet. The dryer's components included a dehydrating chamber, solar heating compartment, biomass heating component, solar PV-driven axial fan component, and exhaust compartment. The functional components of the dryer are shown in the Fig. 4.

The 4 m-long drying chamber of the solar biomass dryer was its main component. Two parts, each measuring 2 m in length, made up this drying chamber. The drying chamber had a semi-circular tunnel shape and was 0.76 m in width. Galvanized iron sheets were used to build the drying chamber's floor, which was painted black to improve heat

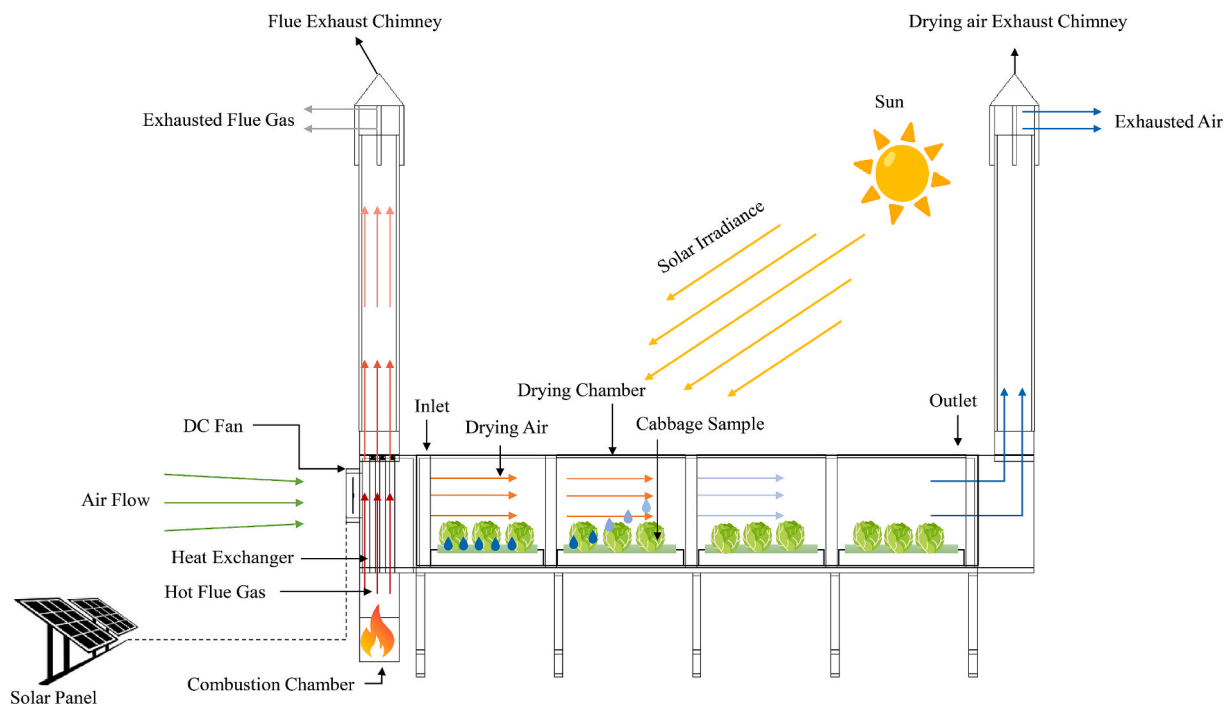


Fig. 2. Schematic Diagram of the GAU Solar-Biomass Hybrid Dryer.

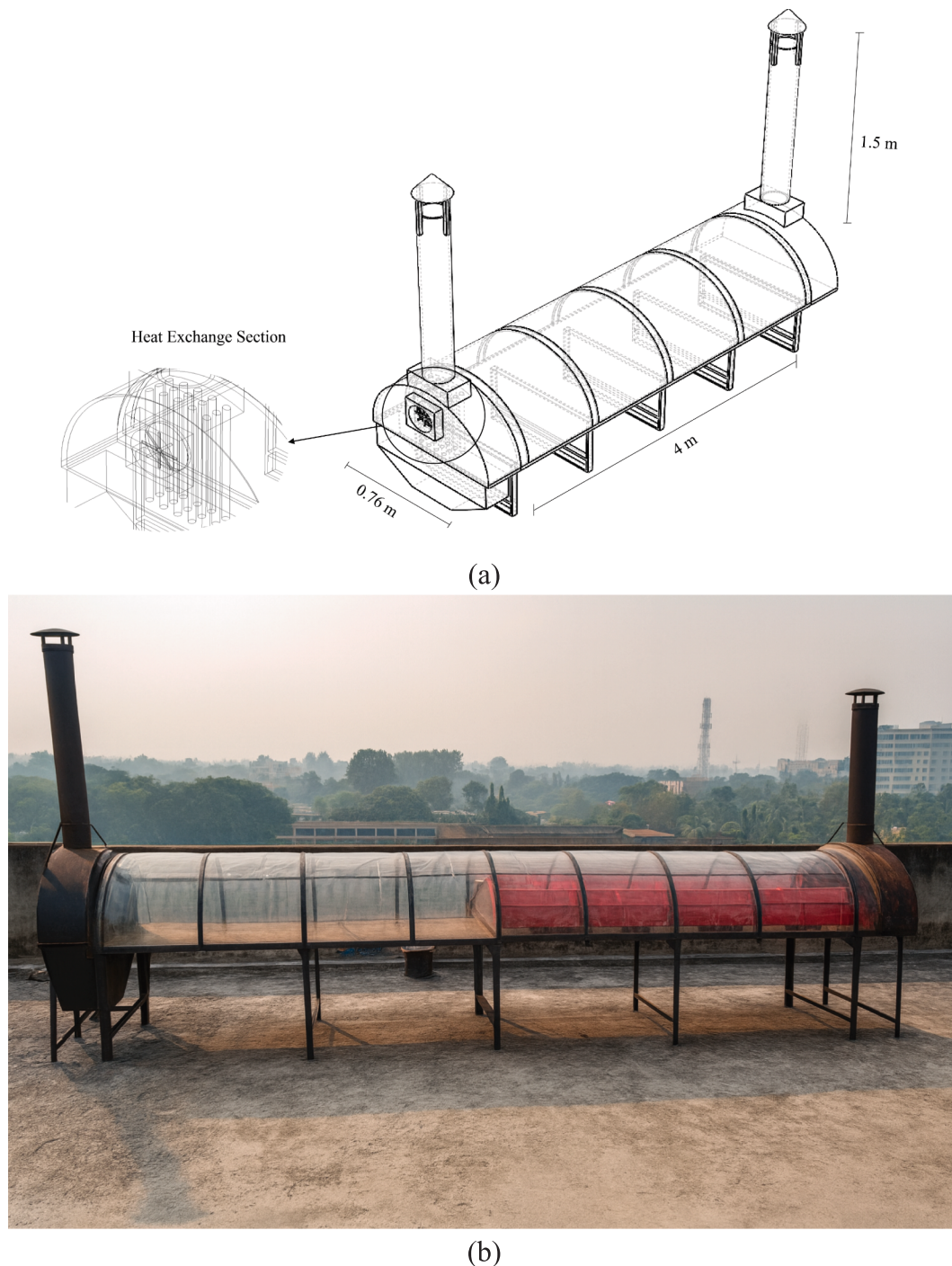


Fig. 3. a) SOLIDWORKS Model, b) Fabricated GAU Solar-Biomass Hybrid Dryer.

absorption. A translucent polyethylene cover was placed on top of the container for drying, allowing sunlight to enter the space. A 2 m-long section near the air entry end of the drying compartment was left blank and painted black. The sunlight entered the drying compartment through a transparent polythene sheet. The black surface generated heat to increase the air temperature inside the chamber.

Biomass combustion was used as an auxiliary heat source to elevate the temperature of the drying air in both the biomass and hybrid drying configurations. A cross-flow heat exchanger, constructed from mild-steel pipes arranged perpendicular to the flue-gas stream, facilitated heat transfer. Ambient air flowed over the heated pipe surface, gained sensible heat by convection, and then entered the drying chamber.

A 10-watt DC fan was connected to enhance the air supply through the drying compartment in order to increase the drying rate. An additional 50-watt PV module was added in parallel connection to the existing one to ensure optimum power supply to the new fan. As the power supply system was increased to a 100-watt solar PV, there was some surplus in electricity generation when there was proper sunshine. A 12-volt 30 A-hour battery was installed in parallel to the PV module to store surplus electricity. The fan was powered by a solar PV module directly when the sunshine produced sufficient electricity, and the battery provided backup power when the sunshine was not enough to run the fan. An exhaust component with a chimney attachment was at the tail end of the chamber. The moist air is exhausted to the environment



Fig. 4. Experimental Setup with all the functional components of GAU Solar-Biomass Hybrid Dryer.

through the chimney. At the precise intersection of the solar-powered heating compartment and the drying compartment, a baffle was built in the upper area of the drying compartment. The purpose of installing this baffle was to enable a steady, consistent airflow across the goods being dried.

The air was heated from three different sources of energy in different configurations: sunlight heating, biomass heating, and a combination of sunlight and biomass heating. In sunlight heating, the dryer tunnel was covered with a transparent polythene sheet. Here, sunlight generated heat on the black surface and heated up the cabbage sample directly. On the other hand, the dryer tunnel was covered with a black opaque polythene sheet to resist the sunlight entering the drying chamber during biomass heating. The air heated up while passing through the heat exchanger. The air heated up from both the sunlight and the biomass heat exchanger in a hybrid heating configuration. The dryer tunnel was covered with a transparent polythene sheet to allow solar heating.

Test sample preparation

Cabbage is a green or purple plant that is cultivated for its dense crowns of leaves, which are harvested as a vegetable crop. It is a very common vegetable enriched in nutritional values and an excellent source of vitamins such as C and E. It was selected as the drying sample not only because of being widely available and consumed but also for its being prone to degradation and sensitive to the post-harvest processes [70,71]. A significant amount of cabbage is wasted each year due to seasonal surplus, poor post-harvest management, shipping, preservation, and lack of food processing facilities. Several studies have shown that the quality of the dried Brassica vegetables is susceptible to drying conditions, especially pretreatments, air temperature and velocity, relative humidity, method of heat supply, and drying time [72–75].

Freshly harvested cabbage was procured and transported to the laboratory. The cabbage was washed, drained, and cut into uniform pieces to control geometric effects on drying. It was chopped into $1\text{ cm} \times 4\text{ cm}$ (about $0.39\text{ in} \times 1.57\text{ in}$) rectangular pieces.

Drying load

Three different batch loads of 4 kg, 8 kg, and 12 kg were prepared for the experiments to incorporate scaling-oriented comparisons. The sliced cabbage was spread on a perforated plastic tray. After that, the trays were placed in the drying compartment. In the drying compartment 12, 16, and 20 trays were used for 4 kg, 8 kg, and 12 kg sample respectively.

For loads of 4 kg, 8 kg, and 12 kg, the mass density in the drying compartment was 2.5 kg/m^2 , 3.7 kg/m^2 , and 4.44 kg/m^2 respectively. The drying compartment and load specifications are given in Table 1.

Instrumentation and data acquisition

All sensors are selected to capture variables required for energy and exergy analysis. Standard instruments were used to measure and record all necessary data. Detailed specifications are presented in Table 2.

Temperature

Ambient air temperature, drying chamber temperature, and biomass heat exchanger outlet temperature were measured at 30-minute intervals using digital thermometers [59,76]. All thermometers were cross-checked against a laboratory-grade reference thermometer in a temperature-controlled water bath to confirm consistency within $\pm 0.2\text{ }^\circ\text{C}$. Prior to experimentation, each thermometer was inspected for response stability and zero-error.

Relative humidity

Relative humidity (RH) of the ambient air and dryer exhaust air was recorded using a digital thermo-hygrometer at the same 30-minute interval [59,76]. The hygrometer was calibrated before the experiment using the saturated salt solution method, ensuring RH stability at 75% RH. The instrument's drift was checked at the start and end of each day to confirm measurement reliability.

Moisture content of cabbage sample

The moisture content of the cabbage samples was determined by the standard gravimetric method [76]. Samples were collected from the dryer at 1-hour intervals and dehydrated in a hot-air oven at $105\text{ }^\circ\text{C}$ for 24 h. The moisture content (mc %) was calculated using:

Table 1

Drying compartment and drying load specifications.

Load, kg	Length of tray (L)	Width of tray (W), m	Area of tray (LW), m^2	No of tray	Total drying area	Density (Load/total area), kg/m^2
4	0.45	0.3	0.135	12	1.62	2.5
8	0.45	0.3	0.135	16	2.16	3.7
12	0.45	0.3	0.135	20	2.7	4.44

Table 2
Technical Specifications of the Measuring Instruments.

Instrument	Parameter	Range	Accuracy
Hukseflux LI-19 Pyranometer	Solar radiation	0–2000 w/m ²	± 1% w/m ²
WT-2 Digital Thermometer	Temperature	–50 – 300 °C	± 1 °C
UNI-T UTA12T Digital Hygrometer	Humidity	10—99%RH	± 5%RH
TESTO-425- Thermal anemometer	Air Velocity	0 to + 20 m/s	±(0.03 m/s)
Digital Balance SF-400D	Weight	0—600 g	± 0.01 g

$$\text{MoistureContent}(mc\%) = \frac{W_2 - W_3}{W_3 - W_1} \times 100$$

Here,

W_1 = Empty crucible weight.

W_2 = Crucible weight containing sample.

W_3 = Crucible weight containing dehydrated sample.

Each moisture content measurement was repeated three times, and the mean value was used for analysis to ensure repeatability and minimize random errors.

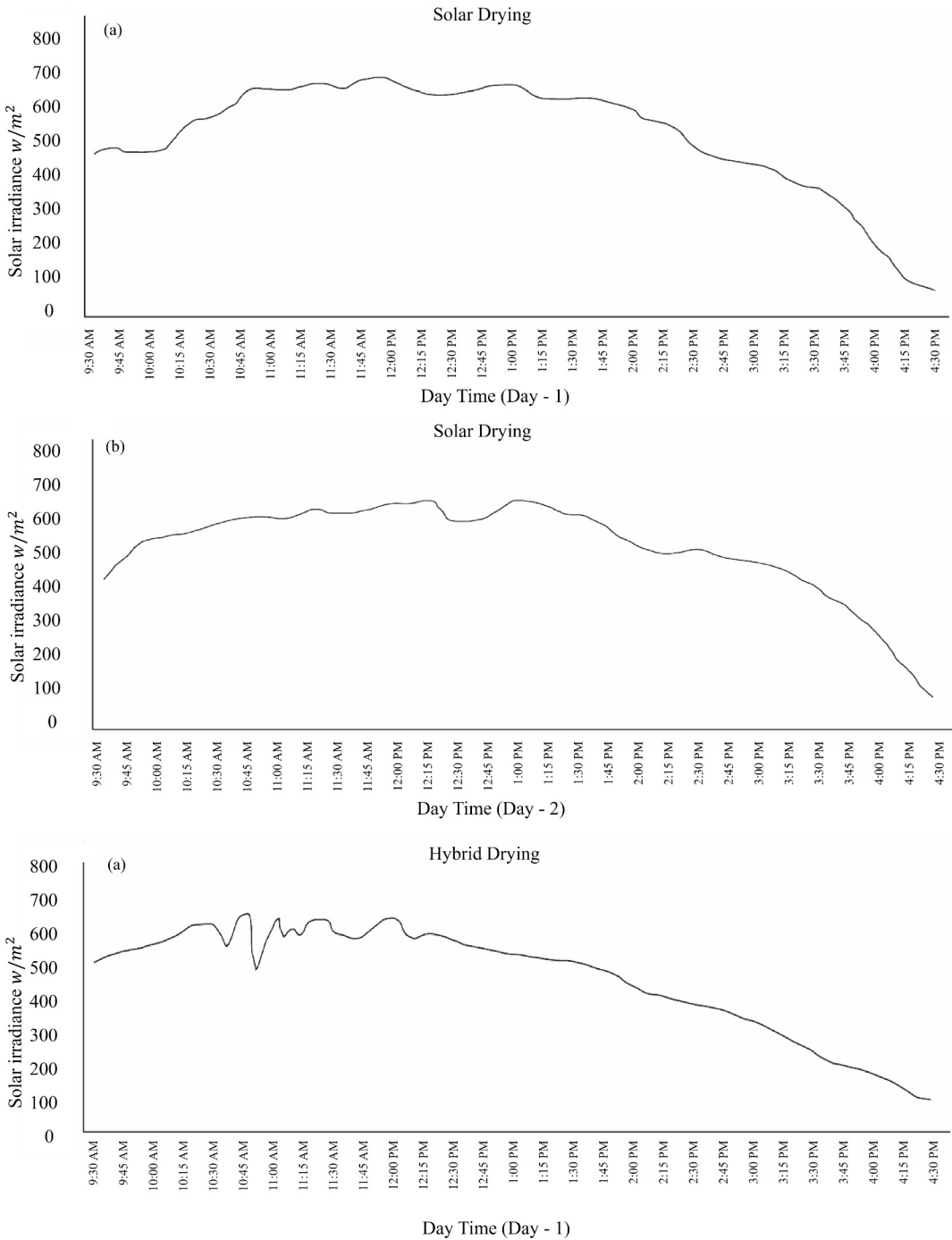


Fig. 5. Solar irradiance profile in Solar Drying (a, b) and Hybrid Drying configuration (c, d).

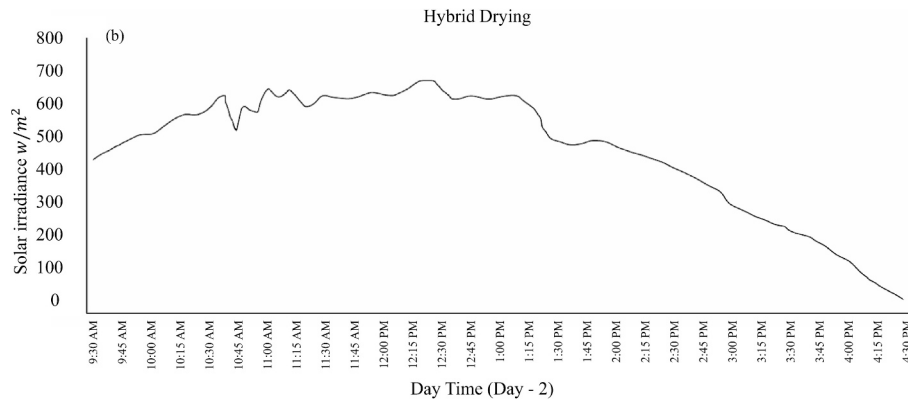


Fig. 5. (continued).

Solar irradiance

Solar irradiance was measured throughout the drying period using a pyranometer (Hukseflux LI-19), and data were logged in W/m^2 at 15 min interval [76]. The device was factory-calibrated, and the calibration certificate was verified prior to experimentation. Recorded irradiance data were downloaded daily after the drying operation ceased. Fig. 5 represents the solar irradiance data logged for the 12 kg load in solar and hybrid drying configurations.

Drying rate. The drying rate of the cabbage was computed utilizing the equation stated

$$\text{Dryin grate (gram moisture removed per hour)} = \frac{\text{Amount of moisture remove (g)}}{\text{Total time}}$$

Measurements were taken using a digital balance with ± 0.01 g accuracy, ensuring precise quantification of weight loss during drying.

Air velocity and airflow

Airflow inside the drying system was measured using a hot-wire anemometer (TESTO-425). The device was zero-calibrated before each measurement session following the manufacturer's guidelines. Air velocity was recorded at multiple locations within the drying chamber to ensure uniform airflow characterization [76].

Experimental procedure

Before each run, the dryer was inspected for leakage, and sensor positions were verified. The airflow system was started and stabilized for 10 min. For solar mode tests, the dryer was operated during daylight hours. For biomass mode and hybrid tests, the biomass unit was ignited and allowed to reach quasi-steady outlet temperature before loading the product. During drying, temperature, RH, air velocity, and solar irradiance were logged continuously, and the product mass was measured at fixed time intervals. Drying continued until the target final moisture criterion was satisfied. After each run, the dryer was allowed to cool, and the chamber was cleaned prior to the next experiment.

Uncertainty and repeatability of the data

Instrument accuracies used in the experiment are provided in Table 2. Based on these accuracies, the associated instrumental uncertainties were $\pm 1 \text{ W/m}^2$ for solar irradiance, $\pm 1^\circ\text{C}$ for temperature, $\pm 5\%$ for RH, $\pm 0.03 \text{ m/s}$ for air velocity, and $\pm 0.01 \text{ g}$ for mass measurement. A detailed overall uncertainty analysis has been presented and compared in the Uncertainty Analysis section.

To verify the reliability of the overall experiment, the drying trials were conducted on two consecutive sunny days with similar

environmental conditions. The drying curves, temperature profiles, and exergy trends showed consistent patterns across both days, confirming the repeatability and robustness of the collected data.

Governing equations

The energy of a system is defined by its ability to perform work. According to the law of conservation of energy, it cannot be created or destroyed, but only transformed from one form to another. Therefore, energy efficiency reflects how effectively a system converts the input energy into the desired work output. In contrast, exergy is defined as the maximum useful work that can be obtained from a system as it comes into equilibrium with a specified environment [77]. Exergy is not bound by conservation law. It can be consumed to perform useful work or can be destroyed due to irreversibilities inherent in the process. These irreversibilities arise as the system moves toward equilibrium with its surroundings, which is why defining a reference environment is essential. Exergy analysis reveals not only how much useful work potential is lost but also where in the system these losses occur. This makes it a powerful diagnostic tool for identifying inefficiencies and room for improvement. That's why exergy and energy analysis are widely applied in studies of thermal systems to enhance performance and sustainability.

In this study, to evaluate the thermodynamic performance of the dryer, both energy and exergy analyses were carried out using a control-volume approach. The drying chamber was treated as a steady-flow system in which hot air enters, interacts with the wet product, and exits as moist air, while moisture is removed from the product as shown in Fig. 6. The models and balance equations used in this study follow the general framework proposed by I. Dincer et al. [78] but are adapted to the specific configuration and operating conditions of the present hybrid solar-biomass dryer.

Assumption and Simplifications

For mathematical formulation, the following assumptions were made, in accordance with widely accepted drying-system thermodynamic models:

1. Each hourly measurement interval is treated as a quasi-steady condition, enabling the application of steady-flow mass, energy, and exergy balances.
2. Air temperatures (T_1 - T_3) and humidity values at the chamber inlet and outlet are averaged to obtain uniform thermodynamic states for analysis.
3. The cabbage undergoes only physical moisture removal, with dry matter treated as inert and chemically non-reactive.

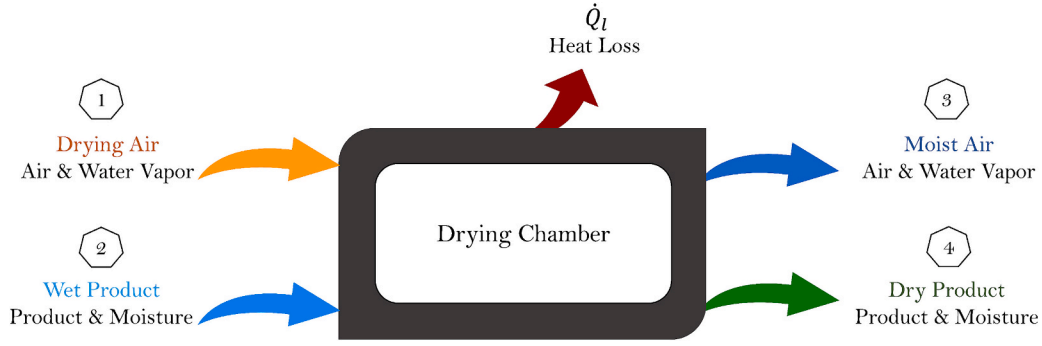


Fig. 6. Schematic Diagram of a Drying Chamber illustrating Mass and Energy Flow.

4. Measured ambient temperature and relative humidity during each interval are used as reference (environmental) conditions for exergy calculations.
5. Only water undergoes evaporation; the dry solid mass remains constant throughout the process.

Mass balance equations

$$\left(\dot{m}_a\right)_1 = \left(\dot{m}_a\right)_3 = \dot{m}_a \quad (1)$$

$$\omega_1 \dot{m}_a + \left(\dot{m}_w\right)_2 = \omega_3 \dot{m}_a + \left(\dot{m}_w\right)_4 \quad (2)$$

$$\left(\dot{m}_p\right)_2 = \left(\dot{m}_p\right)_4 = \dot{m}_p \quad (3)$$

Energy balance equations

$$\dot{m}_a h_1 + \dot{m}_p (h_p)_2 + (\dot{m}_w)_2 (h_w)_2 = \dot{m}_a h_3 + \dot{m}_p (h_p)_4 + (\dot{m}_w)_4 (h_w)_4 + \dot{Q}_l \quad (4)$$

$$\text{Here, } \dot{Q}_l = \dot{m}_a q_1 \text{ is the heat loss from the drying chamber} \quad (5)$$

Enthalpy values h_1 and h_3 of the air at different states can be determined from a psychrometric chart.

Exergy balance equations

$$\dot{m}_a e_1 + \dot{m}_p (e_p)_2 + (\dot{m}_w)_2 (e_w)_2 = \dot{m}_a e_3 + \dot{m}_p (e_p)_4 + (\dot{m}_w)_4 (e_w)_4 + \dot{E}_q + \dot{E}_d \quad (6)$$

Specific exergy of air e_1 and e_3 for state 1 and 3 can be determined from:

$$e_1 = [(C_p)_a + \omega_1 (C_p)_v] (T_1 - T_0) - T_0 \left\{ [(C_p)_a + \omega_1 (C_p)_v] \ln \left(\frac{T_1}{T_0} \right) - (R_1 + R_v \omega_1) \ln \left(\frac{P_1}{P_0} \right) \right\} + T_0 \left\{ (R_1 + R_v \omega_1) \ln \left(\frac{1 + 1.6078 \omega_0}{1 + 1.6078 \omega_1} \right) + 1.6078 \omega_1 R_a \ln \left(\frac{\omega_1}{\omega_0} \right) \right\} \quad (7)$$

$$e_3 = [(C_p)_a + \omega_3 (C_p)_v] (T_3 - T_0) - T_0 \left\{ [(C_p)_a + \omega_3 (C_p)_v] \ln \left(\frac{T_3}{T_0} \right) - (R_3 + R_v \omega_3) \ln \left(\frac{P_3}{P_0} \right) \right\} + T_0 \left\{ (R_3 + R_v \omega_3) \ln \left(\frac{1 + 1.6078 \omega_0}{1 + 1.6078 \omega_3} \right) + 1.6078 \omega_3 R_a \ln \left(\frac{\omega_3}{\omega_0} \right) \right\} \quad (8)$$

Different reference studies incorporated a simplified version of these specific exergy equations [59,79]. Specific exergy of product (e_p), water content (e_w) and exergy flow $\dot{E}_q$ due to heat loss can be obtained from:

$$e_p = [h_p(T, P) - h_p(T_0, P_0)] - T_0 [s_p(T, P) - s_p(T_0, P_0)] \quad (9)$$

$$e_w = h_{fg}(T) - T_0 s_{fg}(T) + T_0 R_v \ln \left(\frac{P_g(T_0)}{x_{v0} P_0} \right) \quad (10)$$

$$\dot{E}_q = \dot{m}_a \left(1 - \frac{T_0}{T_{av}} \right) q_1 = \left(1 - \frac{T_0}{T_{av}} \right) Q_1 \quad (11)$$

Here, T_{av} is the dryer's average surface temperature.

Humidity ratio ω can be obtained from:

$$\omega = 0.662 \frac{RH(T) \cdot P_g(T)}{P - RH \cdot P_g(T)} \quad (12)$$

Energy efficiency

Energy efficiency of the dryer is generally defined as the energy used to evaporate moisture divided by the total energy input.

$$\eta_1 = \frac{\text{Energy used for moisture evaporation}}{\text{Energy input to the system}} \quad (13)$$

$$\eta_1 = \frac{\dot{m}_w (h_{w4} - h_{w2})}{\dot{m}_a (h_1 - h_3)} \quad (14)$$

It can be simplified further as,

$$\eta_1 = \frac{\dot{m}_w \cdot h_{fg}}{\dot{m}_a \cdot C_p (T_1 - T_0)} \quad (15)$$

Exergy efficiency

Exergy efficiency is defined as the ratio of useful work obtained from the system to the maximum possible work potential given to the system. The equation for the drying system as follows:

$$\eta_2 = \frac{\text{Exergy used for useful work (Drying)}}{\text{Exergy of the input drying air}} \quad (16)$$

For the system shown in Fig. 6,

$$\eta_2 = \frac{(\dot{m}_w)_{ev} [(e_w)_3 - (e_w)_2]}{\dot{m}_a e_1} \quad (17)$$

Here, $(\dot{m}_w)_{ev}$ is the moisture removed during the drying.

Uncertainty analysis

Experimental measurements inherently carry uncertainties arising

from instrument accuracy, calibration limitations, environmental fluctuations, and data processing. To ensure reliability and reproducibility of the performance parameters reported in this study, the uncertainty associated with the key parameters and the overall uncertainty in the calculated result (W) have been estimated using the general uncertainty propagation formula presented in Eqn. (18) [23].

$$W = \sqrt{\left(\frac{\partial X}{\partial Y_1} Y_1\right)^2 + \left(\frac{\partial X}{\partial Y_2} Y_2\right)^2 + \dots + \left(\frac{\partial X}{\partial Y_n} Y_n\right)^2} \quad (18)$$

Here, X is the calculated result or output parameter that depends on the input parameters or independent variables Y . W is the total uncertainty in the calculated result X for the overall uncertainty. In the present work, the measured variables include temperature, relative humidity, solar irradiance, and air velocity, all of which directly influence drying behavior and energy-exergy calculations. The associated uncertainties for each parameter, determined from sensor specifications and calibration data, are listed in Table 3.

The combined uncertainty of the experimental measurements was calculated to be 9.95%, which falls within the generally accepted limit of 10% for thermal systems [59]. This indicates that the measured data used for drying kinetics modelling, performance evaluation, and model validation are sufficiently reliable for scientific interpretation and comparison.

To further assess the reliability of the experimental outcomes, a comparison was conducted between the uncertainty level of this study and those reported in similar literature works, as shown in Table 4. The overall uncertainty obtained in the present study is slightly higher than that of the most precise studies, as shown. However, the uncertainty level remains within acceptable bounds and is typical for experiments involving outdoor solar-biomass hybrid thermal environments, where natural variability in meteorological conditions introduces unavoidable fluctuations.

The strong agreement between the present uncertainty value and those previously reported validates the reliability of the measurement procedure and substantiates the robustness of the drying performance results. As the overall uncertainty remains below 10%, the experimental dataset can be considered accurate and suitable for advanced modelling and thermodynamic analysis. Consequently, the model-fitting outcomes and performance indicators derived in this study can be interpreted with high confidence.

Results and discussions

Temperature and humidity behavior

No load conditions

Dryer performance at no load condition was evaluated, recording the inside and ambient relative humidity and temperature. As the temperature inside the chamber increased, the relative humidity decreased. The temperature and relative humidity were observed to be very dependent on sunlight intensity. The highest recorded temperature was 45 °C and 68 °C for sunlight and the combination of sunlight and biomass heating, respectively. When the sunlight and biomass heating were combined, the relative humidity reduced to 6.5%, which indicates an improvement in the drying power.

Table 3
Uncertainty of different drying parameters.

Parameter	Uncertainty
Temperature	± 1.2%
Relative Humidity	± 6.7%
Solar Irradiance	± 7.2%
Air flow	± 1%
Overall	9.95%

Table 4

Comparison of overall uncertainty with reference studies.

Study	Uncertainty	Reference
This study	9.95%	
Debashree et al.	7.7%	[59]
Behra et al.	5.45%	[80]
Choukhi et al.	5%	[68]
Mellalou et al.	5%	[81]
Hao et al.	4.25%	[82]

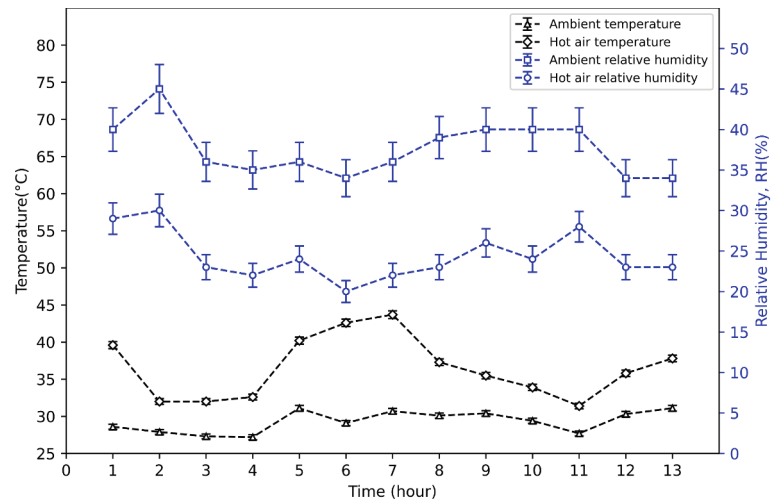
4 Kg load

Measured temperature and humidity in the 4 kg load drying in three modes have been depicted in Fig. 7(a), 7(b), and 7(c) with error bars representing measurement uncertainty based on sensor accuracy. The drying characteristics under sunlight heating were evaluated over a total duration of 13 h, as shown in Fig. 7(a), corresponding to the longest drying time among the three modes for a 4 kg loading. During this period, the maximum solar irradiance reached 821 W/m², with an average irradiance of 614 W/m². The ambient temperature averaged 29.3 °C, while the hot-air temperature inside the drying chamber reached an average of 36.5 °C, yielding a temperature rise of ~ 7.2 °C. This moderate increase is typical in direct solar-heated dryers and is mainly governed by varying cloud cover and solar intensity fluctuations throughout the day. The relative humidity of the inlet air averaged 24%, representing a 12% reduction compared to ambient RH. Such a reduction is essential for enhancing the moisture pickup ability of air. However, the moderate temperature rise limited the moisture diffusivity within the cabbage layers, contributing to the longer total drying time. In biomass heating, as presented in Fig. 7(b), the thermal performance improved significantly. The average hot air temperature observed inside the drying chamber was 58.5 °C, corresponding to an average temperature rise of ~ 26.7 °C compared to ambient air. This substantial thermal lift accelerated internal moisture diffusion and evaporation rates. The average relative humidity of the drying air decreased to 24%, which is 18% lower than the ambient RH. The combination of high temperature and low humidity enhanced the air's moisture-carrying capacity and increased the drying rate. Consequently, the total drying time reduced to 8 h, a reduction of approximately 38% compared to the solar drying mode. This confirms the effectiveness of biomass-based indirect heating as a stabilizing heat source, especially under fluctuating ambient conditions.

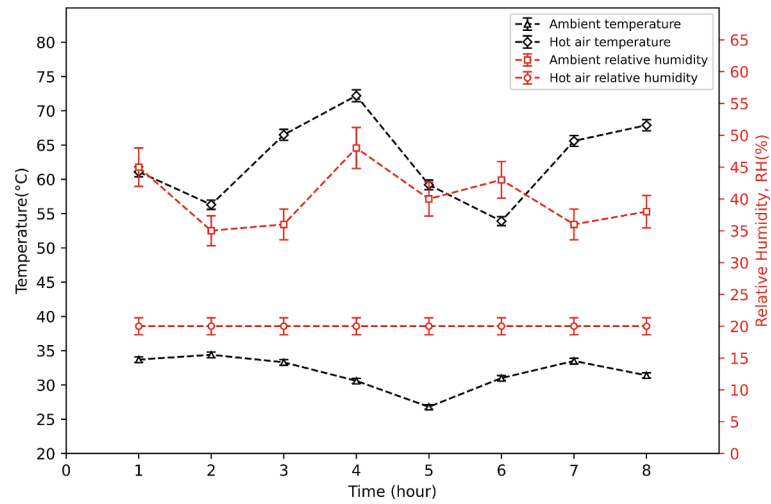
Under hybrid operation, the drying performance improved further, resulting in a total drying time of 7 h as illustrated in Fig. 7(c), which is 1 h shorter than biomass heating mode and 6 h shorter than sunlight heating mode. The average hot air temperature increased to 58.51 °C, starting from an ambient temperature of 30.8 °C, yielding a temperature rise of ~ 27.7 °C. This is the highest effective temperature gain among the three modes. The hybrid system also maintained a consistently low inlet relative humidity of 23%, which enhanced the driving force for moisture removal. The combined heating helped sustain a stable drying environment even during transient solar variations, thus increasing the overall thermal efficiency and reducing exergy destruction in the chamber.

8 Kg load

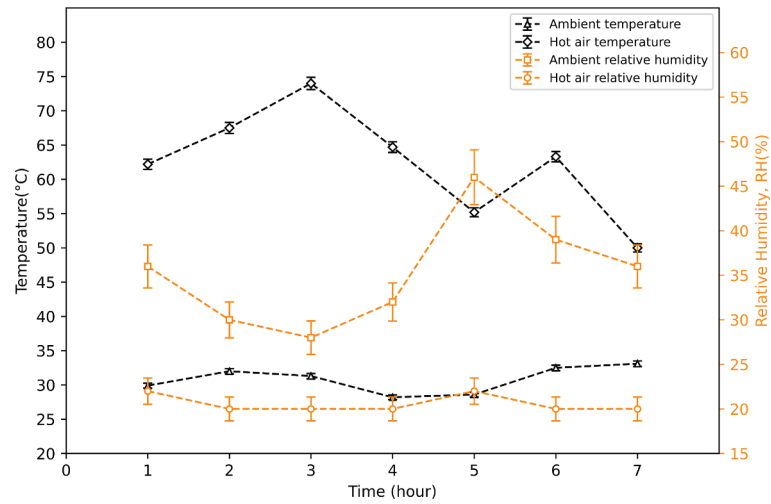
For the 8 kg loading, measured temperature and humidity have been illustrated in Fig. 8(a), 8(b), and 8(c) with error bars. The solar drying mode required a total of 19 h, as shown in Fig. 8 (a), the longest among the three heating modes due to the higher moisture load and the relatively moderate thermal input. The average solar irradiance was 400.8 W/m², with noticeable fluctuations throughout the day, which contributed to variations in ambient and chamber temperatures. The average ambient temperature was 27.5 °C, while the hot-air temperature at the dryer inlet increased to 32.3 °C, giving a modest temperature lift of ~ 4.8 °C. This limited heat gain resulted in a slower internal



(a)

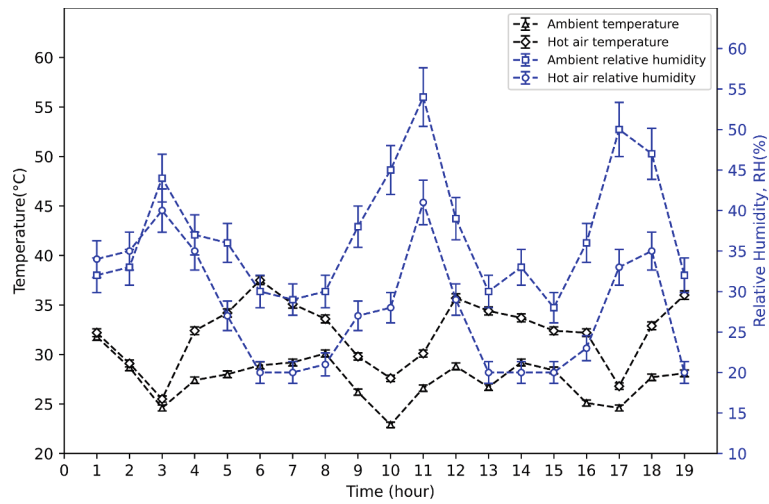


(b)

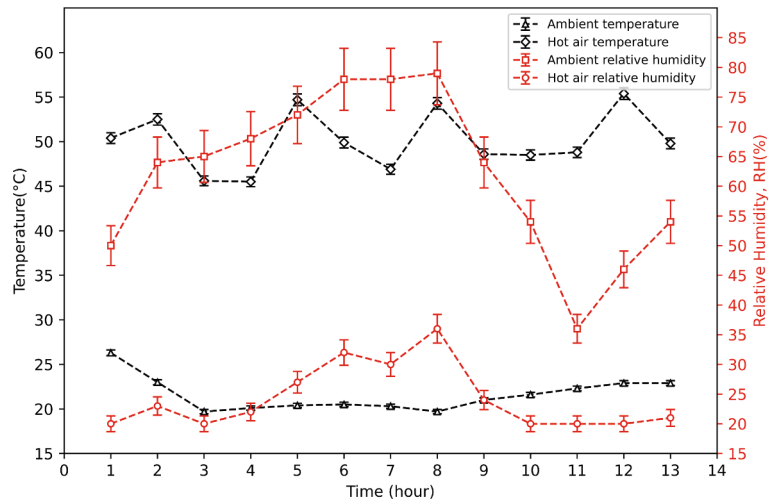


(c)

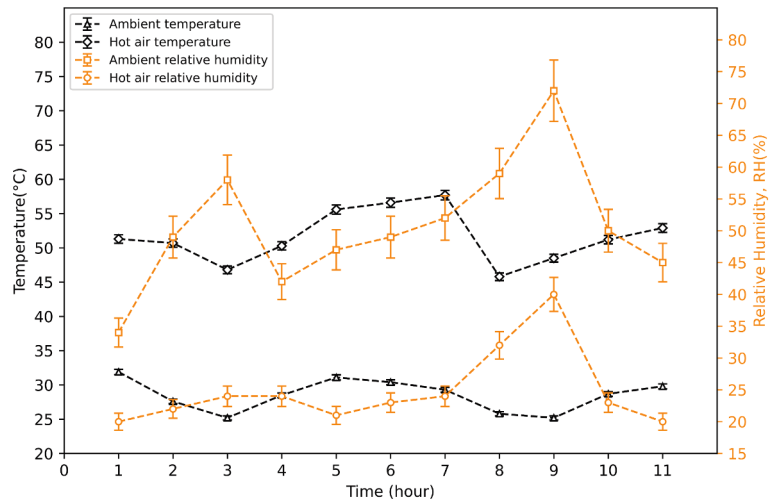
Fig. 7. Temperature and Humidity profile for 4 kg load in (a) Solar Drying, (b) Biomass Drying, (C) Hybrid Drying with error bars for measurement uncertainty.



(a)



(b)



(c)

Fig. 8. Temperature and Humidity profile for 8 kg load in (a) Solar Drying, (b) Biomass Drying, (C) Hybrid Drying with error bars for measurement uncertainty.

moisture migration within the thicker cabbage layers. The relative humidity of the ambient air was consistently high, and the drying process reduced the inlet RH by 9%, resulting in an average inlet RH of $\sim 27.7\%$. Although this reduction improved the moisture absorption capability of the air, the modest temperature rise was insufficient to accelerate evaporation significantly, causing the long total drying time.

Under biomass heating, the hot-air temperature inside the dryer reached an average of 49°C as demonstrated in Fig. 8 (b), increasing from a much cooler ambient temperature of 21.5°C , corresponding to a thermal lift of $\sim 27.5^\circ\text{C}$. Such an increase greatly enhanced the evaporation rate of moisture from the cabbage. The ambient relative humidity averaged 62.15% , while the RH at the dryer inlet reduced to 24.23% , representing a $\sim 38\%$ reduction. This substantial drop in RH increased the drying potential of the air and influenced faster moisture transfer from the cabbage surface. As a result, the total drying time was reduced to 13 h, showing a 31% improvement compared to sunlight drying. The high and stable heat supplied by the biomass unit compensated for the fluctuating sunlight and improved the uniformity of the drying conditions. Hybrid heating further enhanced the performance by combining the steady thermal output from biomass combustion with the additional contribution of solar radiation. The total drying time was reduced to 12 h, the lowest among the three modes, as the profile shown in Fig. 8(c).

During hybrid operation, the average air temperature at the dryer inlet increased to 51.5°C , starting from an ambient value of 28.5°C , giving a temperature rise of $\sim 23^\circ\text{C}$. This strong thermal input accelerated both surface evaporation and internal moisture diffusion. The relative humidity also dropped substantially from an ambient RH of 50.6% to an inlet RH of 24.8% , indicating improved moisture-picking ability of the drying air. The synergy between solar heating and biomass combustion created a more stable and higher-temperature drying environment, which contributed to the observed reduction in drying time.

12 Kg load

For the highest loading condition, measured temperature and humidity have been presented in Fig. 9(a), (b), and (c) with error bars. The drying process using sunlight as the sole energy source required 20 h, which is expected due to the increased product mass and moisture content. The average solar irradiance was 533.5 W/m^2 , higher than during the 8 kg experiment, although fluctuations in solar intensity caused noticeable variations in the temperature profile. The ambient temperature averaged 27°C , while the hot-air temperature inside the drying chamber reached 32.3°C , providing a modest temperature increment of $\sim 5.3^\circ\text{C}$ as the profile shown in Fig. 9 (a). This limited thermal lift slowed down the internal moisture diffusion in the dense cabbage layers, contributing to prolonged drying. The relative humidity at the dryer inlet decreased to an average of 27.6% , showing adequate moisture absorption capability. However, due to the relatively moderate temperature rise, the solar mode remained insufficient to handle the large product load efficiently, resulting in the longest drying duration of all tested modes.

In Biomass heating, the ambient temperature during the experiment averaged 22°C , yet the hot-air temperature at the inlet rose dramatically to 51.7°C , representing a substantial temperature gain of $\sim 29.7^\circ\text{C}$ as shown in Fig. 9(b). This increase created a strong temperature gradient, enhancing both surface evaporation and internal moisture diffusion. The relative humidity of the drying air showed a sharp reduction from 50.5% ambient RH to 23.9% at the dryer inlet, resulting in a $\sim 26.6\%$ drop. Such a high-temperature, low-humidity environment is ideal for reducing drying time, especially for thick vegetable layers. Consequently, the total drying time decreased to 18 h, demonstrating a considerable improvement over sunlight heating. The stability of biomass heating enabled more consistent moisture removal despite fluctuations in ambient conditions. Hybrid heating produced the best drying performance for the 12 kg load as illustrated in Fig. 9(c). With

the synergistic effect of solar and biomass heating, the total drying time was reduced to 13 h, corresponding to a 35% reduction compared to biomass heating and a 55% reduction relative to sunlight heating. The average ambient temperature during hybrid operation was 27.2°C , and the hot-air temperature increased to 51.2°C , yielding a temperature lift of $\sim 24^\circ\text{C}$. Although slightly lower than the biomass mode temperature increment, the hybrid mode benefited from the combined and more stable thermal input over time. The relative humidity decreased from 35.14% ambient RH to 21.42% at the inlet, indicating enhanced drying air quality. The consistent high-temperature and low-humidity environment led to efficient evaporation and significantly faster drying. The hybrid system again demonstrated its capability to maintain stable heat input during solar variations, making it particularly effective for large loads such as 12 kg.

Moisture content reduction and drying rate

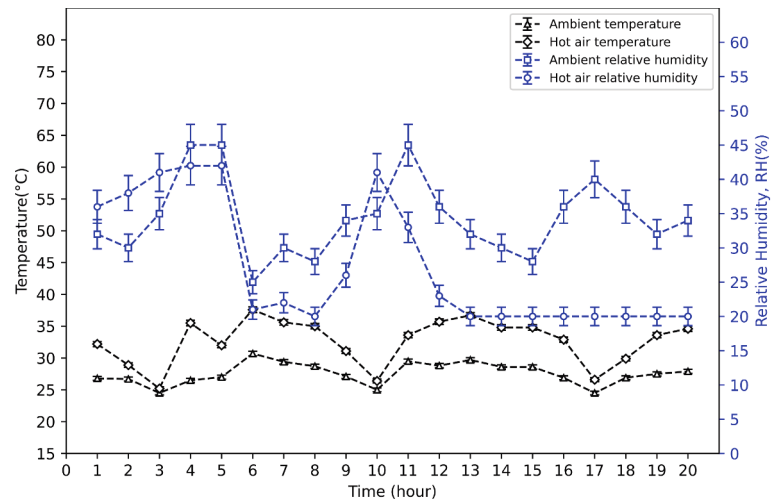
Across all loading conditions, moisture content decreased from 90% to 8% as presented in Fig. 10 (a-c), with drying time and drying rate strongly governed by the heating mode and product loading. The moisture removal rate for all drying loads in three configurations is presented in Fig. 10 (d). For the 4 kg batch, sunlight, biomass, and hybrid heating resulted in drying rates of 261.53, 425.00, and 485.71 g/h, respectively. Increasing the load to 8 kg raised the drying durations, and the corresponding drying rates were 353.68, 523.20, and 566.67 g/h. The 12 kg batch, which had the highest moisture load, required higher drying hours, producing drying rates of 492, 600, and 766.15 g/h for sunlight, biomass, and hybrid modes, respectively. The hybrid configuration consistently achieved the shortest drying time and the highest drying rates in all drying loads. It is attributed to its ability to maintain elevated inlet-air temperatures and substantially lower humidity compared with sunlight drying, which remained sensitive to fluctuations in solar irradiance. Biomass heating provided more stable thermal input but exhibited lower performance than hybrid operation due to the absence of supplemental solar heat during peak intensity periods. The results indicate that higher inlet-air temperatures and reduced humidity significantly enhance the air's moisture-carrying capacity, accelerating internal moisture diffusion and surface evaporation. All tests were conducted while maintaining conditions that preserved acceptable food quality and prevented overdrying. The final dried product sample is presented in Fig. 11.

Biomass consumption

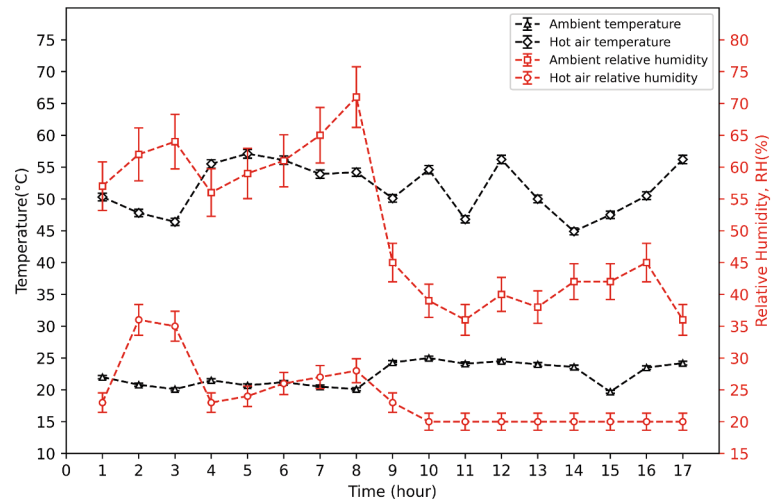
Locally available wood was used as the biomass energy source for the drying process. The consumption of biomass was carefully recorded for different drying modes and sample loads as illustrated in Fig. 12, providing insight into the relative efficiency of the energy sources. In the experiment, drying the 4 kg sample with biomass heating mode required around 33 kg of wood, whereas the combined use of sunlight and biomass reduced the requirement to 25 kg. This trend was consistent with larger sample sizes. For the 8 kg sample, the biomass mode consumed 46 kg compared to 40 kg for the hybrid mode. In the case of the 12 kg sample, the biomass method required 59 kg of biomass, while the combined heating method used only 45 kg.

Exergy and energy efficiency

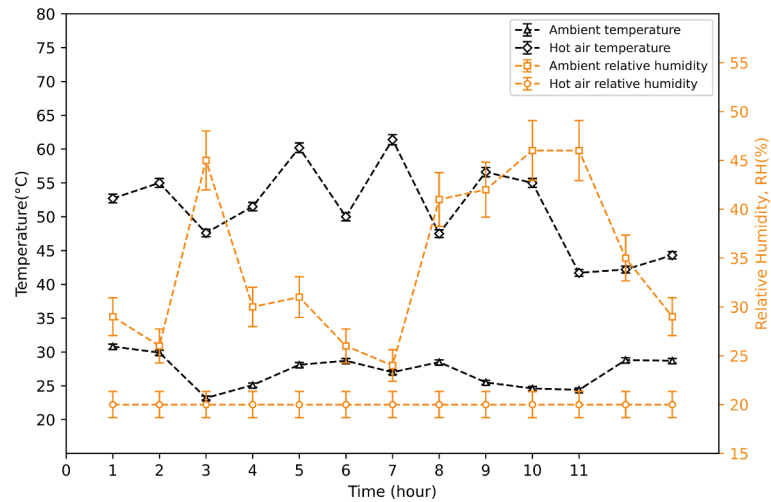
The calculated energy-exergy efficiencies and exergy usage-destruction for the 8 kg load in different drying configurations are presented in Table 5 and Table 6, respectively. These performance indicators reflect the effectiveness of the system in converting input energy into useful work and the extent of irreversibilities in the drying process, respectively. Energy efficiency for the different configurations shows a noticeable variation in drying performance. Sunlight heating achieved an energy efficiency of 38%, while biomass heating showed a



(a)



(b)



(c)

Fig. 9. Temperature and Humidity profile for 12 kg load in (a) Solar Drying, (b) Biomass Drying, (C) Hybrid Drying with error bars for measurement uncertainty.

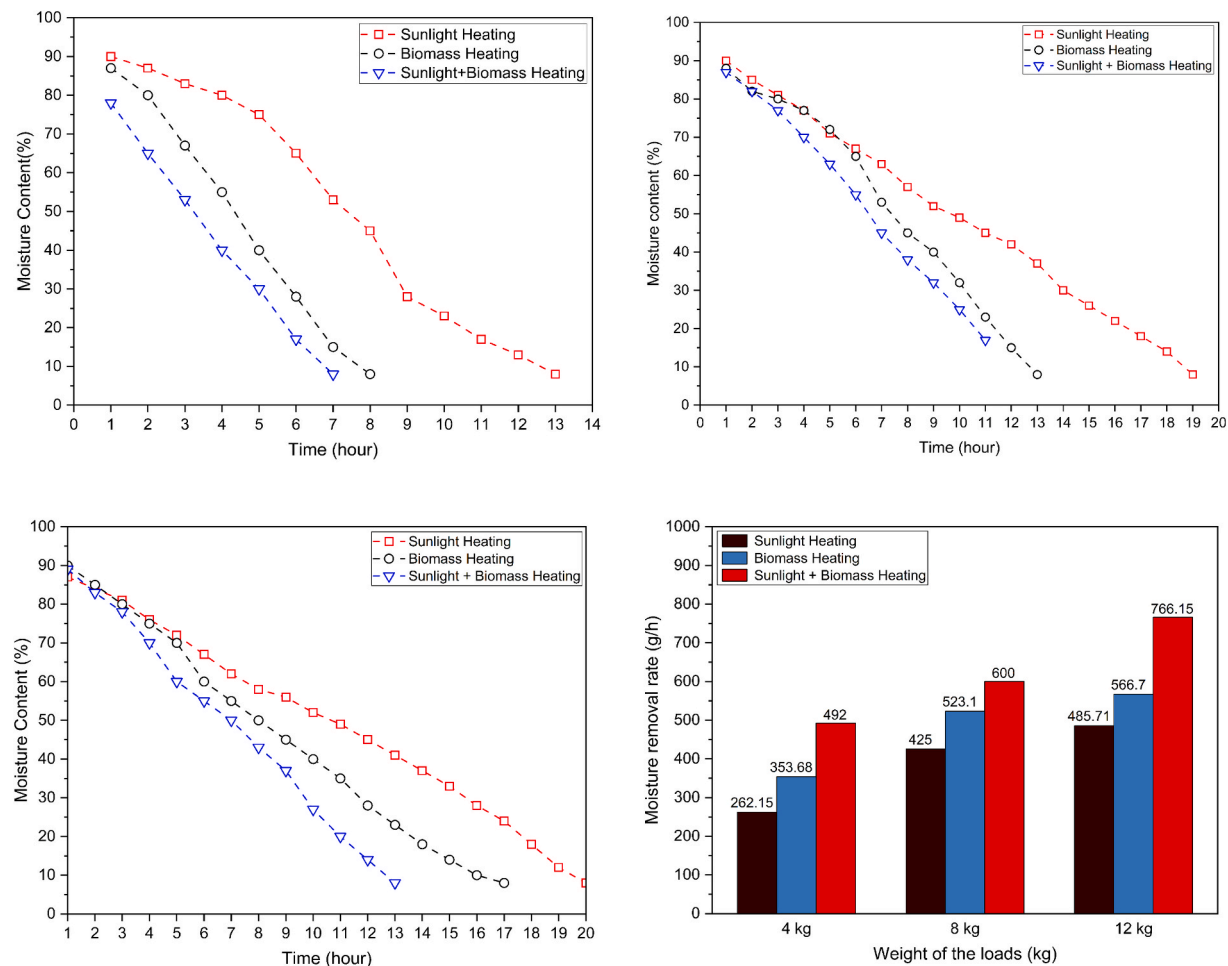


Fig. 10. Moisture content reduction (a-c) and removal rate (d) in 4 kg, 8 kg, and 12 kg loads upon sunlight, biomass, and sunlight-biomass combined heating.



Fig. 11. Final dried product in hybrid drying configuration.

lower efficiency of 33%. The hybrid configuration exhibited the highest energy efficiency at 50%. These results indicate the enhanced thermal performance of the hybrid system, which benefits from the synergy of both heating sources to improve heat utilization. The reduction in drying time from 19 h (sunlight heating) to 12 h (hybrid heating) correlates with the increased energy efficiency, demonstrating the efficiency gains from combining renewable energy sources with biomass.

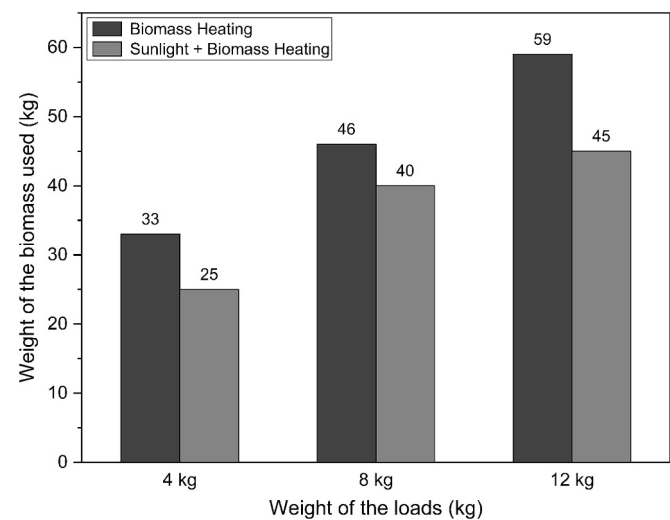


Fig. 12. Biomass consumption in different loading conditions and heating configurations.

efficiency further corroborates the results observed for energy efficiency. The sunlight heating mode showed an exergy efficiency of 37%, while biomass heating had a similar exergy efficiency of 33%. The hybrid system again outperformed the others with an exergy efficiency of 45%, emphasizing the reduced exergy destruction when combining

Table 5

Energy and Exergy efficiencies for dryer drying 8 kg load.

Configuration	Drying Time (hr.)	Energy Efficiency (η_1)	Exergy Efficiency (η_2)
Sunlight Heating	19	38%	37%
Biomass Heating	13	33%	33%
Sunlight-Biomass Hybrid	12	50%	45%

Table 6

Exergy used and destroyed in three drying configurations.

Configuration	Exergy Used (MJ)	Exergy Destroyed (MJ)
Sunlight Heating	20.23	33.91
Biomass Heating	91.96	167.54
Sunlight-Biomass Hybrid	83.93	107.42

solar and biomass heating.

The exergy used and destroyed during the process, as given in Table 6, also reflects the efficiency differences. The hybrid system consumed 83.93 MJ of exergy and destroyed 107.42 MJ, compared to 167.54 MJ destroyed in biomass heating. The significant reduction in exergy destruction in the hybrid system highlights its potential for more sustainable drying operations, as it utilizes energy more effectively and reduces losses. The exergy loss and destruction in the drying process are primarily attributed to the inherent inefficiencies in energy conversion, heat transfer, and system irreversibilities. In the case of sunlight heating, the energy from the sun is not always optimally absorbed by the system due to variations in solar radiation, leading to lower exergy efficiency. Similarly, the biomass heating system, while relatively efficient, involves combustion processes that produce significant waste heat and emissions, further contributing to exergy destruction. The hybrid configuration mitigates some of these inefficiencies by combining the complementary strengths of both energy sources. Solar energy is harnessed during peak sunlight hours, while biomass heating compensates for periods of low solar radiation, reducing reliance on a single energy source and improving overall system performance. However, even in the hybrid system, some exergy destruction is inevitable due to the irreversible nature of energy transfer and conversion processes, particularly in the biomass combustion and heat exchange stages.

The overall improved drying performance in the hybrid configuration can be attributed to continuous and stable heat supply. This reduces the fluctuations in energy availability that affect systems reliant on a single energy source. The system benefits from the synergy of both heating sources, which optimizes heat utilization and minimizes energy losses. The use of forced convection and shell-and-tube heat exchangers in this configuration further enhances heat transfer efficiency, accelerating the drying process. This combination of energy sources and advanced heat transfer techniques allows the hybrid system to reduce drying time and achieve higher efficiencies compared to the individual sunlight or biomass systems.

Table 7 compares the drying times and efficiencies of this study with those of similar studies found in the literature. Compared to the study by D. Behera et al. [59] this study demonstrates higher energy and exergy efficiencies. The hybrid system in this study (12 h) outperforms many solar or solar-biomass hybrid systems reported in the literature, such as those by Rajesh et al. [83] and Madhankumar et al. [84], which showed longer drying times and lower efficiencies. This suggests that the combination of forced convection, shell and tube heat exchangers, and hybrid heating in this study leads to a more efficient and effective drying process, supporting its potential for broader application, especially in regions with variable sunlight availability.

Table 7

Comparison of experimental results of this study with literatures.

Reference	Test Scenarios	Configuration	Drying Temperature	Drying Efficiency (%)
This Study	Cabbage 4 kg (13 h, 8 h, 7 h) 8 kg (19 h, 13 h, 12 h) 12 kg (20 h, 17 h, 13 h)	Solar, Biomass, Solar Biomass Hybrid dryer with forced convection and shell and tube heat exchanger.	Sunlight – 32 Biomass – 50 Hybrid – 52	Sunlight – 38 Biomass – 32 Hybrid – 50
D. Behera et al. [59]	Tomato Slices, Potato Slices, Banana, Bitter Guard 5 kg (5 h, 4 h, 3 h)	Absorbing plate with fins, used PCM for paraffin wax and Silica for drying agent	69	33.61
Madhankumar et al. [84]	Momordica Charantia 2 kg (10.5 h)	PCM and Fin are integrated with the indirect solar dryer	–	19.4
Surendra et al. [85]	Indian Jujube, Date palm	PV Hybrid solar dryer with PCM	–	18
Rajesh et al. [83]	Elephant apple	Forced convection solar dryer with PCM	60	12
A.K. Bhardwaj et al. [86]	Valeriana Jatamansi herb 50 gm (5 days)	Indirect Solar Dyer with PCM	51.22	
C.D. Constantino et al. [87]	Tithonia diversifolia Gray 2.5 kg	Hybrid Solar dryer with Flat Plate collector	59	

Drying model

The drying behaviour of the agricultural product under sunlight, biomass, and hybrid heating configurations was quantitatively analysed through seven well-established thin-layer drying models as listed in Table 8. The experimental moisture ratio (MR) data exhibited a characteristic falling-rate trend across all heating modes, indicating that internal moisture diffusion rather than surface evaporation governed the overall drying mechanism. To identify the most suitable mathematical representation of this behaviour, the MR data for each system were fitted to the models. The goodness-of-fit was assessed using the coefficient of determination (R^2), root mean square error (RMSE), and reduced chi-square (χ^2). The fitted parameters and associated statistical indices are summarized in Table 9.

Table 8

Different drying models.

No	Model	Equation	References
1	Newton	$mr = e^{-kt}$	[88]
2	Henderson and Pabis	$mr = Ae^{-kt}$	[89]
3	Logarithmic	$mr = Ae^{-kt} + B$	[90]
4	Page	$mr = e^{-kt^n}$	[91]
5	Two-term	$mr = Ae^{-k_1t} + Be^{-k_2t}$	[92]
6	Midilli and Kucuk	$mr = Ae^{-kt^n} + Bt$	[93]
7	Wang and Singh	$mr = 1 + At + Bt^2$	[94]

Table 9

Drying model parameters and fitting criteria.

Model	Drying Type	Model Constants	R ²	RMSE	χ ²
Lewis	Sunlight	k = 0.07040	0.9078	0.08322	0.00731
	Biomass	k = 0.08773	0.8073	0.1306	0.01848
	Hybrid	k = 0.10056	0.8472	0.11373	0.01411
Henderson & Pabis	Sunlight	A = 1.17767, k = 0.08697	0.9561	0.05739	0.00368
	Biomass	A = 1.26376, k = 0.12001	0.8988	0.09467	0.01059
	Hybrid	A = 1.26455, k = 0.13651	0.9364	0.07338	0.00646
Logarithmic	Sunlight	A = 39.21452, k = 0.00129, B = -38.16605	0.9988	0.00951	0.00011
	Biomass	A = 4034.32280, k = 0.00002, B = -4033.17680	0.9818	0.04012	0.00209
	Hybrid	A = 2829.77560, k = 0.00003, B = -2828.65505	0.9963	0.0177	0.00042
Page	Sunlight	k = 0.01364, n = 1.67047	0.9914	0.02534	0.00072
	Biomass	k = 0.00473, n = 2.36464	0.9935	0.02395	0.00068
	Hybrid	k = 0.01164, n = 2.05956	0.9955	0.01947	0.00045
Two-term	Sunlight	A = 0.80743, k ₁ = 0.08697, B = 0.37024, k ₂ ≈ 0	0.9561	0.05739	0.00417
	Biomass	A = 0.76152, k ₁ = 0.12001, B = 0.50224, k ₂ = 0.12001	0.8988	0.09467	0.01295
	Hybrid	A = 0.86532, k ₁ = 0.13651, B = 0.39922, k ₂ = 0.13652	0.9364	0.07338	0.00808
Midilli-Kucuk	Sunlight	A = 1.41655, k = 0.29990, n = 0.01438, B = -0.04938	0.9988	0.00946	0.00011
	Biomass	A = 1.00042, k = 0.00388, n = 2.29283, B = -0.01190	0.9966	0.01745	0.00044
	Hybrid	A = 1.04159, k = 0.01383, n = 1.70366, B = -0.02464	0.9987	0.01068	0.00017
Wang-Singh	Sunlight	A = -0.04071, B = -0.00039	0.996	0.01732	0.00034
	Biomass	A = -0.02627, B = -0.00359	0.9904	0.02913	0.001
	Hybrid	A = -0.04221, B = -0.00293	0.9904	0.02851	0.00098

Across all drying techniques, the Midilli-Kucuk model consistently provided the best fit, yielding the highest R² values (0.9987–0.9988) and the lowest RMSE (0.00946–0.01068) and χ² values (1.1 × 10⁻⁴–1.7 × 10⁻⁴). The strong performance of this semi-empirical model is due to its combined exponential-linear structure, which effectively captures the nonlinearity associated with the rapid early-stage moisture removal and the diffusion-controlled slow drying period. This behaviour aligns with trends reported for solar and mixed-mode dryers in literature, where the Midilli-Kucuk model frequently outperforms both empirical and theoretical models due to its flexibility in representing complex heat-mass transfer interactions under fluctuating drying air conditions.

The Logarithmic model emerged as the second-best performer for all drying scenarios. Its R² values remained above 0.9818 and improved to 0.9988 for sunlight heating, with remarkably low RMSE across the configurations (0.0095–0.0177). The additional constant term in this

model allows adjustment of curve curvature, which is particularly beneficial under sunlight and hybrid conditions where temperature and humidity vary dynamically. The high accuracy of this model confirms the presence of multi-stage drying behaviour, where external convection dominates initially but later transitions to internal moisture diffusion resistance. The Page model also demonstrated strong predictive capability (R² = 0.9914–0.9955), especially for the biomass and hybrid systems. The exponent n > 1 obtained for these configurations reflects the accelerated decrease in MR during the early drying phase, likely induced by higher and more stable inlet temperatures of biomass and hybrid heating compared to sunlight alone. These findings reaffirm that the Page model adequately represents falling-rate drying of biological materials without requiring complex mechanistic formulations.

The Henderson-Pabis and Two-term models provided moderate agreement with experimental data (R² = 0.8988–0.9561). Although they captured the overall drying trend, their lower performance relative to Midilli-Kucuk and Logarithmic models suggests limited capability in describing the curvature changes occurring between rapid surface evaporation and diffusion-controlled periods. Similarly, the Lewis model, being purely exponential, showed the weakest performance (R² = 0.8073–0.9078). This confirms that single-parameter models are insufficient to characterize the nonlinear drying behavior observed under mixed-mode heating systems, where varying temperature and humidity profiles influence mass transfer mechanisms. Interestingly, the Wang-Singh model exhibited relatively high R² values (0.9904–0.9960), but its quadratic structure lacks physical interpretability, and RMSE and χ² values were consistently higher than those of the best-performing models. Thus, despite a good numerical fit, it is not ideal for the mechanistic interpretation of drying kinetics.

Economic analysis

One of the important aspects of the adoption of a renewable energy-based hybrid drying system is its economic feasibility, especially in resource-constrained grid regions where conventional grid electricity is barely available and is not cost-effective. This economic analysis evaluates the feasibility of the solar-biomass hybrid dryer from economic aspects through different matrices, including capital cost, capital recovery factor, maintenance expenses, system lifetime, maintenance cost, and energy cost. The analysis utilizes experimental data and estimates data based on comparisons with similar hybrid drying systems from the literature [95]. All cost equations used in this analysis are adopted from the reference study [55].

The yearly cost of the solar biomass hybrid dryer can initially be determined from the capital cost and the capital recovery factor of the solar biomass hybrid dryer.

$$\text{Initial Year Cost} = \text{Capital Cost} \times \text{Capital Recovery Factor} \quad (19)$$

The capital cost of the hybrid dryer is assumed to be 165 \$, and the Capital Recovery Factor is given by,

$$\text{Capital Recovery Factor} = \frac{i(1+i)^n}{i(1+i)^n - 1} \quad (20)$$

where *i* and *n* are the annual interest rate and the lifetime of the solar biomass hybrid dryer, respectively. For this study, the lifetime is taken as 5 years. The annual interest rate is taken 5%, and the maintenance cost has been estimated to be 15% of the initial year cost [95]. The yearly salvage value is given by,

$$\text{Yearly Salvage Value} = \text{Salvage Value} \times \text{System's Sinking Fund} \quad (21)$$

where the salvage value after 5 years of the lifetime of the hybrid dryer is assumed to be 41 \$ and the system's sinking fund is given by

$$\text{System's Sinking Fund} = \frac{i}{(1+i)^n - 1} \quad (22)$$

The total yearly cost of the hybrid dryer can be found by

$$\begin{aligned} \text{Total Yearly Cost} = & \text{Initial Year Cost} + \text{Yearly Maintenance Cost} \\ & + \text{Yearly Biomass Cost} - \text{Yearly Salvage Value} \end{aligned} \quad (23)$$

The yearly biomass cost is estimated to be 45 \$. The yearly useful energy from solar radiation can be determined by

$$\begin{aligned} \text{Yearly solar useful energy} & \left(E_{\text{useful,solar}} \right) \\ = & \text{Average total solar incident radiation on drying chamber} \left(\frac{W}{m^2} \right) \\ & \times \text{drying chamber area} (m^2) \times \text{efficiency of the system} (\eta_1) \\ & \times \text{average number of sunny days per year} \\ & \times \text{hours of operation per day} \end{aligned} \quad (24)$$

The average total solar incident radiation in BD is taken to be 600 W/m² [96]. The approximate number of sunny days in Bangladesh per year is assumed to be 220 days, and the number of operating hours per day for this case is taken as 7 h. The yearly useful energy from biomass can be determined by,

$$E_{\text{useful,biomass}} = m_{\text{biomass}} \times \text{LHV} \times \eta_{\text{system}} \times \text{operating days a year} \quad (25)$$

Where the mass of biomass is approximately 10 kg/per day, and LHV is taken as 14 MJ/kg, and η_{system} for a typical biomass furnace in dryers is taken 80% [97,98].

Finally, the cost of useful energy can be calculated from the equation. (26).

$$\text{Useful Energy Cost} = \frac{\text{Yearly cost}}{\text{Yearly solar useful energy} + \text{yearly biomass useful energy}} \quad (26)$$

For solar mode, the cost of useful energy can be given by:

$$\text{Useful Energy Cost} = \frac{\text{Yearly Cost}}{\text{Yearly Solar Useful Energy}} \quad (27)$$

And for biomass mode, the cost of useful energy can be given by:

$$\text{Useful Energy Cost} = \frac{\text{Yearly Cost}}{\text{Yearly Biomass Useful Energy}} \quad (28)$$

Comparative Economic Analysis presented in Table 10 shows that the system solely relying upon solar is the most expensive one, considering its cost per unit of energy, due to high capital investment and yielding very low energy. While the biomass configuration is most effective, it produces less useful energy than the solar biomass hybrid configuration, and attempting to produce the same amount of useful energy, it increases the carbon footprint. Both the solar and biomass configuration relies on a single source of energy, which doesn't give energy security in off-grid remote places. On the other hand, the hybrid configuration

Table 10
Cost assessment in three drying configurations.

System	Annual Useful Energy (kWh)	Annual Cost (\$)	Cost per Unit Energy (\$/kWh)
Solar	778.38	37	0.04753
Solar + Biomass	7,622.78	82	0.01075
Hybrid			
Biomass	6,844.44	45	0.00657

delivers the highest overall useful energy while remaining only slightly more expensive than the biomass mode and approximately 77% cheaper than the solar heating configuration. In addition, it integrates two complementary energy sources, providing mutual backup, reducing dependence on environmental variability, and lowering the carbon footprint through the incorporation of renewable solar energy. As the additional cost of transitioning from biomass to hybrid mode is minimal, the hybrid configuration can offer a more reliable and sustainable long-term drying solution.

Environmental impact assessment

The hybrid solar-biomass dryer substantially reduces greenhouse gas emissions compared to conventional biomass drying methods by combining solar energy. This results in lower biomass usage as given in Table 11. Thus, it reduces carbon footprints. Greenhouse gas emission through biomass combustion (considering wood as biomass) can be found using wood emission factors of 1.593 kg CO₂/kg dry wood, 0.5 g CH₄/kg, and 0.03 g N₂O/kg. [99,100] For the biomass configuration, to produce 6,844.44 kWh of useful energy, approximately 11.14 kg of wet wood (8.91 kg dry based on 20% moisture) is to be burnt daily. For 220 operating days yearly, this results in the emissions of 3,122.53 kg CO₂, 0.98 kg CH₄, and 0.059 kg N₂O. The global warming potential factors (CH₄ × 28 and N₂O × 265) can be used to find out the total annual CO₂ equivalent (CO₂e) emission, which is approximately 3,165.55 kg CO₂e. In contrast, the hybrid solar-biomass system reduces biomass consumption by about 40% for the same annual useful energy output of 6,844.4 kWh. Following the same approach used previously, equivalent (CO₂e) emission for hybrid solar biomass is found to be approximately 1,899.33 kg CO₂e per year, which is much less than that of the biomass

drying configuration [101].

The integration of renewable solar energy with biomass in a hybrid dryer lowers the dependency on biomass and mitigates CO₂ emissions. Along with lowering carbon emissions, it can reduce post-harvest food loss, local air pollution, and the demand for forest biomass resources at the same time, which can help to save trees in rural areas.

Conclusion and future recommendations

This study experimentally evaluated the drying performance and thermodynamic characteristics of a hybrid solar-biomass dryer for cabbage under tropical humid conditions using three loading levels and three heating configurations. The work offers novel insights into hybrid drying for high-moisture vegetables by integrating solar and biomass heat through a shell-and-tube heat exchanger and forced convection. An approach not previously reported for humid tropical climates. The system performance was assessed through drying time, drying rate, energy efficiency, exergy efficiency, exergy destruction, thin-layer drying

Table 11
Carbon emission comparison in biomass and hybrid heating mode.

System	Useful Energy (kWh/year)	Annual CO ₂ e Emissions (kg)	CO ₂ e Intensity (kg/kWh)
Biomass (wood)	6,844.4	3,165.55	0.46
Solar-biomass hybrid	6,844.4	1,899.33	0.28

models, economic evaluation, and carbon emission comparison. Primary findings are as follows:

- Hybrid drying mode consistently achieved the best thermal performance, with energy and exergy efficiencies of 50% and 45%, significantly higher than standalone sunlight drying (38%, 37%) and biomass drying mode (33%, 33%).
- Drying times were reduced to 7–13 h under hybrid drying operation, representing up to 40–50% faster drying than sunlight drying.
- Drying rates were highest in hybrid heating, reaching 766.15 g/h for the 12 kg load.
- Midilli-Kucuk model best represented the drying behavior, with $R^2 \approx 0.9988$ and the lowest RMSE and χ^2 values, confirming diffusion-controlled drying under fluctuating inlet conditions.
- Hybrid system delivered the highest annual useful energy (7,622.78 kWh), at only 0.01075 \$/kWh, making it 77% cheaper than solar drying and slightly more costly than biomass drying.
- Hybrid drying reduced carbon emissions by 39% compared to biomass drying mode, with a CO₂e intensity of 0.28 kg/kWh.

Although the hybrid solar-biomass system demonstrated strong thermodynamic, economic, and environmental performance, it was not possible to address all the variability of real-world factors such as biomass fuel quality and seasonal humidity and temperature variation over the year. Moreover, the crucial system components, such as heat exchanger design, collector area, and airflow system, can be further optimized, and stability in long term usage can be assessed. Future research should therefore focus on testing the dryer under diverse climatic and operational conditions, performing multi-objective optimization of design parameters to minimize exergy destruction and cost, integrating advanced sensor-based and IR 4.0 control systems for real-time regulation of drying parameters to support commercial-scale deployment and adaptation for different agricultural products and regional contexts.

CRediT authorship contribution statement

Sifat Abdul Bari: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Mohtasim Fuad:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Md. Shahed Mahamud:** Writing – original draft, Methodology, Formal analysis, Data curation. **Arafat Ahmed Bhuiyan:** Writing – review & editing, Supervision, Methodology. **Md. Ahiduzzaman:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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